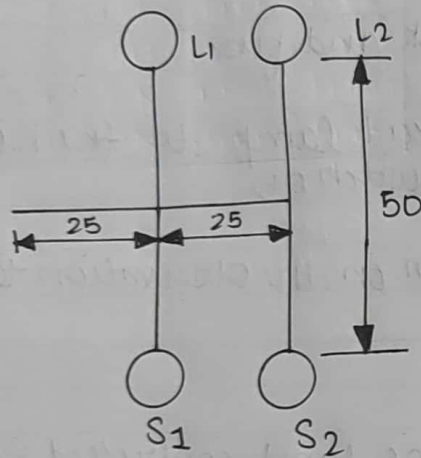
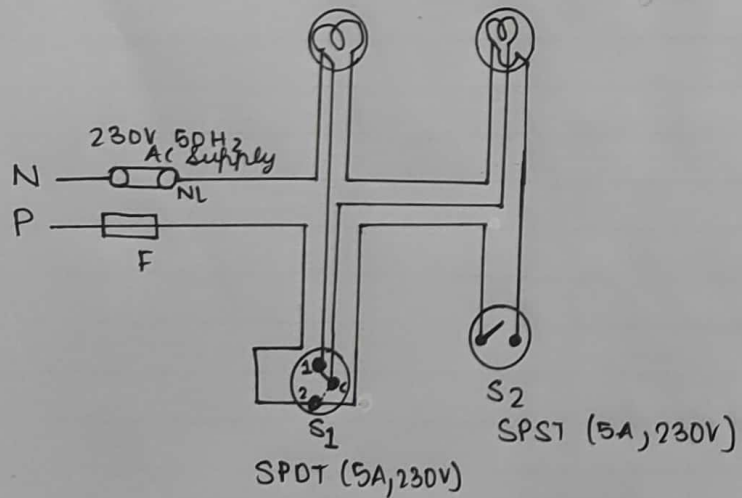


LAYOUT DIAGRAM



CIRCUIT DIAGRAM



HOSPITAL WIRING

28

AIM

Wire up a circuit to control two lamps by two switches to get dim light and bright light in open conduit wiring system as per layout

DESIGN

$$\text{Power, } P = VI \cos \phi$$

$$I = P/V$$

$$P = 120W$$

$$V = 230V$$

$$\begin{aligned} \text{Load current} &= 1.5 \times \text{load current} \\ &= 1.5 \times 0.52 \\ &= \underline{0.783A} \end{aligned}$$

OBSERVATION TABLE

S ₁	S ₂	L ₁	L ₂
C-1	OFF	DARK	DARK
C-2	OFF	DIM	DIM
C-1	ON	BRIGHT	BRIGHT
C-2	ON	BRIGHT	DARK

MATERIALS REQUIRED

S.No	Items	Specification	Rating	Quantity
1	Switch	SPDT single pole double throw	5A, 230V	1
		SPST single pole single throw	5A, 230V	1
2	Lamp	Incandescent	60W, 250V	1
3	Conduit	20mm PVC		150cm
4	Copper wire	1 sq mm		740cm
5	Junction	20mm 3 way PVC		1
		20mm 4 way PVC		1
6	Round Block	3 inch PVC		2
7	Switch	3x2" PVC		2
8	Clamp/ Saddle	20mm Metal		6
9	Fuse Unit	Kitt Kat (binding materials)	16A	1

MATERIALS REQUIRED

No.	Item	Specification	Quantity
1	Switch box	10A	1
2	Conduit	1/2"	10m
3	Wire	1/2"	10m
4	Wire	3/4"	10m
5	Wire	1"	10m
6	Wire	1 1/4"	10m
7	Wire	2"	10m
8	Wire	3"	10m
9	Wire	4"	10m
10	Wire	5"	10m
11	Wire	6"	10m
12	Wire	8"	10m
13	Wire	10"	10m
14	Wire	12"	10m
15	Wire	14"	10m
16	Wire	16"	10m
17	Wire	18"	10m
18	Wire	20"	10m
19	Wire	22"	10m
20	Wire	24"	10m

10. Fuse wire

Tin & Lead alloy

2A

10cm

11. Neutral link

Porcelain base with copper segment

16A

1

12. Insulation tape

1

13. Screw

1/2"

12

3/4"

6

1 1/4"

2

TOOLS REQUIRED

Screw driver, line tester, combination pliers, ball pen hammer, fork, nose pliers, wire strippers.

PROCEDURE

1. Draw the circuit diagram and check whether it is correct
2. Fix the conduit to the board using saddle of appropriate diameter
3. Wire the circuit as per the circuit diagram.
4. Fix the switch box and give its connections

5. Fix the round block to the base board
6. Fix the loop holder on top of the round block and give its connections
7. Attach neutral link and fuse
8. Add the incandescent lamp to the holder & turn the switch on.
9. Record observation on the observation table.

RESULT

A circuit with 2 lamps controlled by 2 switches to get dim light and bright light in open conduit wiring system as per layout was wired up and an estimation was prepared for the same.